LESSON 01 / 06

A matrix is a linear machine

An $m \times n$ matrix $A = (a_{ij})$ has m rows and n columns; its entry a_{ij} is in row i, column j. Multiplication by a column vector combines rows with coordinates:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad x = \begin{pmatrix} u \\ v \end{pmatrix}, \quad Ax = \begin{pmatrix} u + 2v \\ 3u + 4v \end{pmatrix}.$$

INTUITION

Rows act; columns record

The row i of A computes the ith output as a dot product. The columns are the outputs of the coordinate basis vectors: Ae_j is column j.

Shape check before multiplying

$A_{m \times n} B_{n \times p}$ exists and has shape $m \times p$; its entry (i, j) is $\sum_{k=1}^{n} a_{ik} b_{kj}$. Inner dimensions must agree. Addition needs identical shapes. Matrix multiplication is associative and distributive, generally **not commutative**.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix} \neq \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix}.$$

COMMON TRAP

Order and zero traps

AB defined does not imply BA is defined. $AB = 0$ does not force $A = 0$ or $B = 0$; singular matrices can annihilate nonzero vectors. Never cancel a matrix factor unless invertibility is established.

Identity, powers and transpose

$I_n A = A = A I_m$ when dimensions fit. $A^0 = I$ for square A; powers require a square matrix. $(AB)^T = B^T A^T$ (reverse order), $(A + B)^T = A^T + B^T$.

WORKED EXAMPLE

Quick transpose check

If $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, then $A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$, now 3×2. Transposing swaps row/column labels, not the entries' signs.

Named structures

Square: $n \times n$. Diagonal: off-diagonal entries zero. Scalar: λI. Symmetric: $A^T = A$; skew-symmetric: $A^T = -A$, hence diagonal entries are zero over real numbers. For any real square A:

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}$$

is the unique symmetric + skew-symmetric decomposition.

KEY RESULT

Transpose identities

$(A^T)^T = A$; $(AB)^T = B^T A^T$; $(A^{-1})^T = (A^T)^{-1}$ when inverse exists. The last follows by transposing $AA^{-1} = I$.

LESSON 02 / 06

The determinant measures signed area / volume

For $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det A = ad - bc$. The columns span a parallelogram; $|\det A|$ is its area scale, while the sign records orientation. In dimension n, determinant is the signed volume scale.

For a 3×3 matrix expand along a row/column: delete row i, column j to get minor M_{ij}; cofactor $C_{ij} = (-1)^{i+j} M_{ij}$. Then $\det A = \sum_j a_{ij} C_{ij}$. The checkerboard signs begin $+, -, +$.

$$\det\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix} = 1(50-48) - 2(40-42)$$

$$+3(32-35) = -3.$$

SHORT DERIVATION

Why the cofactor signs alternate

In the permutation formula, exchanging two columns reverses orientation. Deleting row i and column j leaves the same smaller determinant; the parity of the row/column moves contributes $(-1)^{i+j}$.

Row operations: distinguish the three

1. Swap two rows: determinant changes sign.
2. Multiply a row by c: determinant multiplies by c.
3. Add c times one row to another: determinant unchanged.

Same rules hold for columns. Thus if rows are equal/proportional, determinant is zero; a common row factor can be pulled out. Triangular determinant is product of diagonal entries.

COMMON TRAP

Scaling the whole matrix

For an $n \times n$ matrix, $\det(cA) = c^n \det A$ because each of the n rows scales. It is not merely $c \det A$ unless $n = 1$.

Structural identities

$$\det(A^T) = \det A, \qquad \det(AB) = \det A \det B$$

$$\det(A^{-1}) = \frac{1}{\det A}.$$

The inverse identity assumes $\det A \neq 0$. A is singular iff $\det A = 0$; geometrically its columns collapse into a lower-dimensional space.

WORKED EXAMPLE

Parameter test

$A = \begin{pmatrix} k & 1 \\ 2 & k \end{pmatrix}$ is invertible iff $k^2 - 2 \neq 0$, i.e. $k \neq \pm\sqrt{2}$. At either excluded value the columns are dependent and the homogeneous system has a nonzero solution.

KEY RESULT

Fast evaluation

Before expanding, look for a sparse row/column; repeated/proportional rows; row additions that create zeros; triangular form. Keep track of swaps and row scalings explicitly.

LESSON 03 / 06

Cofactors assemble the adjoint

For square A, $C_{ij} = (-1)^{i+j} M_{ij}$ and $\operatorname{adj} A = (C_{ij})^T$: **transpose the cofactor matrix**. Cofactor expansion gives

$$A \operatorname{adj} A = \operatorname{adj} A A = (\det A) I.$$

Off-diagonal entries cancel in pairs; diagonal entries are cofactor expansions of $\det A$.

KEY RESULT

Inverse criterion and formula

A^{-1} exists exactly when $\det A \neq 0$. Then $A^{-1} = \dfrac{\operatorname{adj} A}{\det A}$. If determinant vanishes, no inverse: the adjoint formula is not a workaround.

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\operatorname{adj} A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0.$$

WORKED EXAMPLE

Compute and verify

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, determinant -2, adjoint $\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$, hence $A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$. Check by multiplication: $AA^{-1} = I$.

Useful inverse laws (only when defined)

$(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$, $(A^{-1})^{-1} = A$. Also $\det(A^{-1}) = 1/\det A$. To solve $AX = B$, multiply **on the left**: $X = A^{-1}B$; preserve order.

COMMON TRAP

Adjoint versus cofactor matrix

The adjoint is the transpose of the cofactor matrix. A symmetric matrix can hide this distinction because its cofactor matrix may also be symmetric; verify the definition instead of relying on that coincidence.

A practical 3x3 calculation

For $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$, cofactors give $\det A = 4$ and $\operatorname{adj} A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}$. Therefore $A^{-1} = \frac{1}{4} \operatorname{adj} A$; the quick check is $A \operatorname{adj} A = 4I$.

LESSON 04 / 06

Encode equations as $AX = B$

For n linear equations in n unknowns, coefficient matrix A, unknown vector X, constants B. If $\det A \neq 0$, there is exactly one solution, $X = A^{-1}B$.

Cramer's rule

Replace column i of A by B to form A_i. Then $x_i = \det A_i / \det A$. This requires square A and nonzero determinant.

WORKED EXAMPLE

Two equations

$2x + y = 5$, $x - y = 1$. Here $A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ and $\det A = -3$. Replacing columns gives determinants -6, -3, hence $(x, y) = (2, 1)$. Substitute back to check both equations.

WORKED EXAMPLE

Three unknowns

For $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$, coefficient determinant is 7. Replacing columns by $(6, 3, 2)^T$ gives determinants $7, 14, 21$, so $(x, y, z) = (1, 2, 3)$. Direct substitution checks all three equations.

COMMON TRAP

Cramer's rule does not settle singular cases

If $\det A = 0$, do not divide. The system may have no solution or infinitely many; compare the equations/ranks or row-reduce the augmented matrix.

Homogeneous systems

$AX = 0$ always has $X = 0$. A nonzero solution exists iff A is singular (for square A), since $AX = 0$ has a nontrivial null vector exactly when columns are dependent.

WORKED EXAMPLE

Parameter condition

$\begin{pmatrix} \lambda & 1 \\ 1 & \lambda \end{pmatrix} X = 0$ has nonzero solutions iff $\lambda^2 - 1 = 0$, so $\lambda = \pm 1$. For other real λ, only $X = 0$.

General consistency and rank

For m equations, n unknowns, compare coefficient rank $r(A)$ and augmented rank $r([A|B])$:

- unequal ranks: no solution;
- equal rank $r = n$: unique solution;
- equal rank $r < n$: infinitely many solutions (over $\mathbb{R}$).

The last case has $n - r$ free parameters. Rank is the number of pivots after row reduction, equivalently the largest order of a nonzero minor.

SHORT DERIVATION

Why ranks decide consistency

Row operations preserve the equation solution set. An echelon system has one pivot equation per pivot; a row $[0 \cdots 0 \mid c]$, $c \neq 0$, is a contradiction. Without contradiction, each non-pivot variable is free.

KEY RESULT

Syllabus boundary

JEE matrix methods: determinants up to order 3, adjoint/inverse, and Cramer's rule for 2- or 3-variable systems. Rank language is a useful explanatory lens; do not let it replace the required determinant conditions.

LESSON 05 / 06

Symmetric and skew-symmetric

Real square A is symmetric when $A^T = A$; skew-symmetric when $A^T = -A$. In the latter, $a_{ii} = -a_{ii}$, so every diagonal entry is zero. Every real square matrix splits uniquely:

$$A = S + K, \quad S = \frac{A + A^T}{2}, \quad K = \frac{A - A^T}{2}.$$

WORKED EXAMPLE

Split a 2×2 matrix

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $S = \begin{pmatrix} 1 & 5/2 \\ 5/2 & 4 \end{pmatrix}$ and $K = \begin{pmatrix} 0 & -1/2 \\ 1/2 & 0 \end{pmatrix}$. Check $S^T = S$, $K^T = -K$, and $S + K = A$.

Orthogonal matrices preserve length

Real square Q is orthogonal if $Q^TQ = QQ^T = I$. Thus $Q^{-1} = Q^T$ and columns (also rows) are orthonormal. For every vector x, $\|Qx\|^2 = x^TQ^TQx = \|x\|^2$: rotations/reflections preserve lengths and angles. Taking determinants: $(\det Q)^2 = 1$, so $\det Q = \pm 1$.

$$Q = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is a rotation; check orthogonality using $\cos^2\theta + \sin^2\theta = 1$.

COMMON TRAP

Transpose is not inverse in general

$A^T = A^{-1}$ only for orthogonal A. Symmetry $A^T = A$ alone does not imply orthogonality or invertibility.

Three algebraic families

Idempotent: $P^2 = P$ (projection; eigenvalue intuition only in extension). Involutory: $R^2 = I$, hence $R^{-1} = R$. Nilpotent: $N^k = 0$ for some positive integer k; nilpotent matrices are singular when positive-dimensional.

WORKED EXAMPLE

A projection

$P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ has $P^2 = P$. Acting on $(x, y)^T$, it maps to $(x + y, 0)^T$; applying it again changes nothing.

KEY RESULT

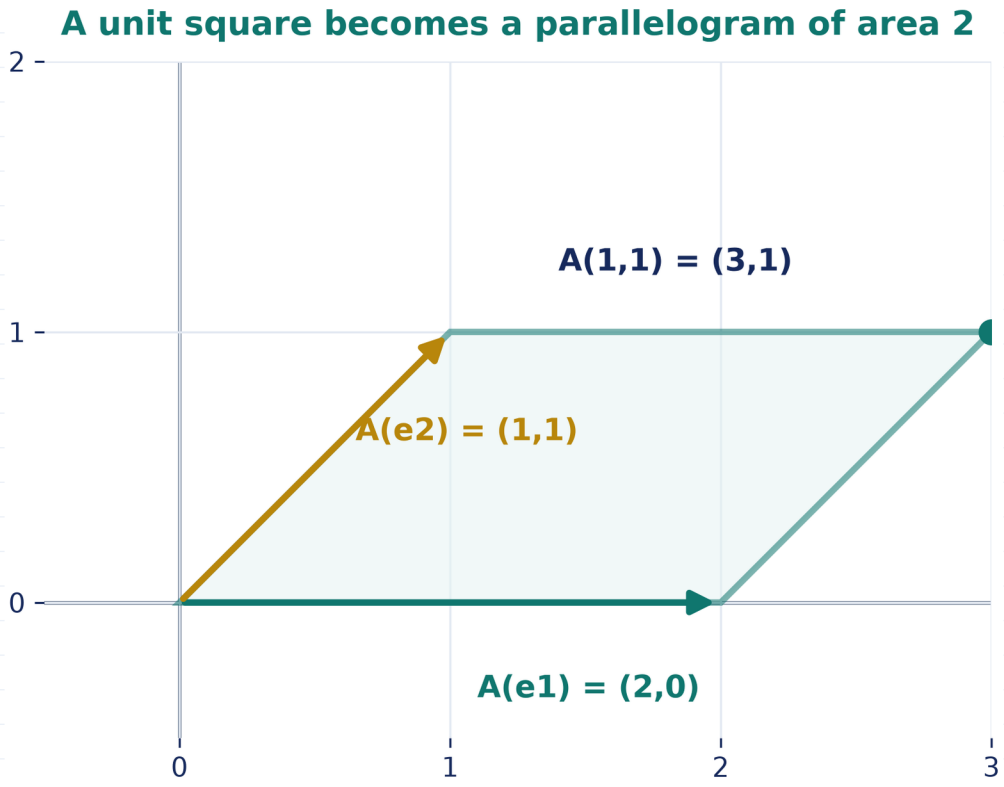
Test the definition

To classify a given matrix, compute the defining product: A^T, A^2, or AA^T. A visual pattern is not a proof.

LESSON 06 / 06

Columns are images of the basis vectors

For $A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, the first column is $Ae_1 = (2, 0)^T$ and the second is $Ae_2 = (1, 1)^T$. The whole unit square maps to the parallelogram spanned by those two columns.



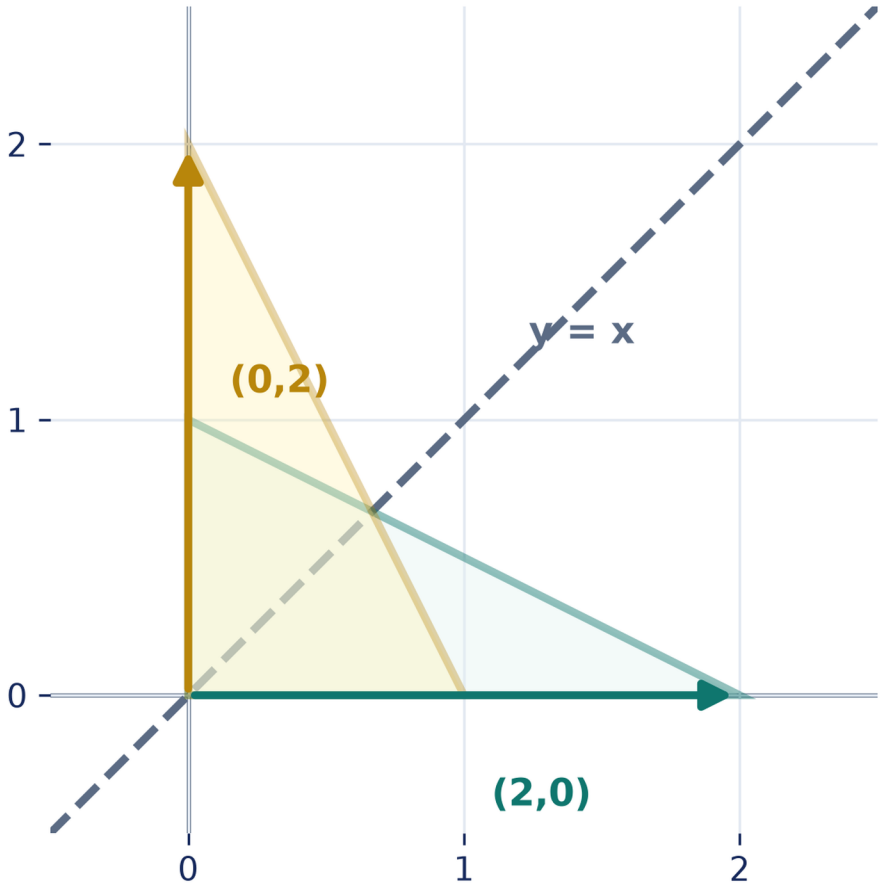
READ THE FIGURE
The determinant is 2*1 - 1*0 = 2. The parallelogram has base 2 and height 1, so its area is also 2: the determinant is the signed area scale.

If the determinant is zero, the two columns are dependent and the parallelogram collapses to a segment or point. The matrix then cannot be inverted because distinct inputs can share an output.

Determinant sign records orientation

Swapping coordinates is reflection in $y = x$. Areas stay the same but orientation reverses, so the determinant is -1. More generally, $|\det A|$ is the area or volume scale, while the sign tells whether orientation is preserved or reversed.

Swapping coordinates reverses orientation



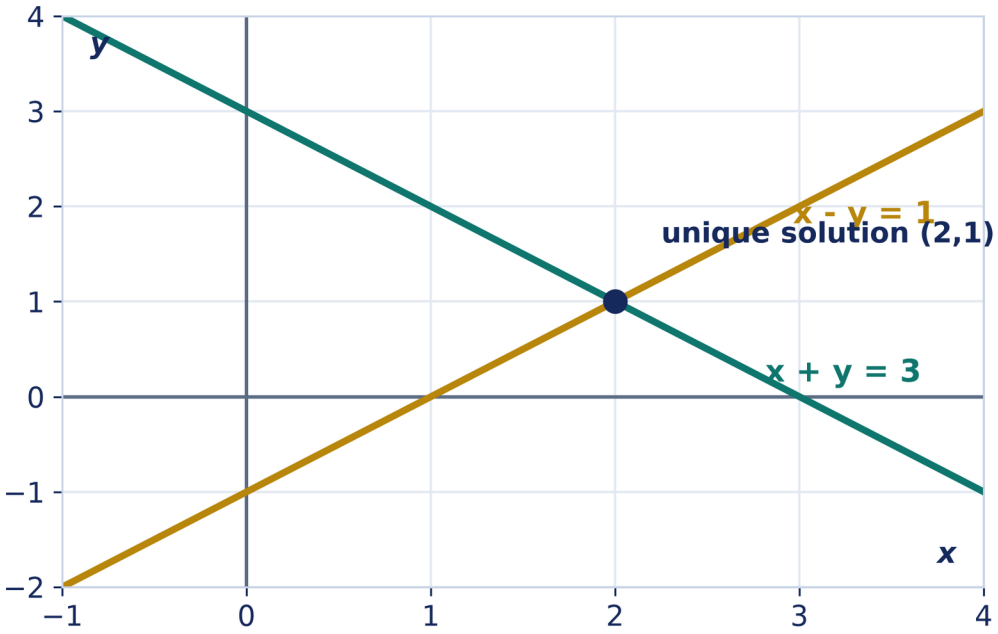
READ THE FIGURE
The matrix [[0,1],[1,0]] reflects in y = x. Areas stay the same but orientation reverses, so its determinant is -1.

For a 3×3 matrix, the same interpretation uses signed volume. The identities $\det(AB) = \det A \det B$ and $\det(A^T) = \det A$ become plausible respectively as successive volume scaling and preservation of the parallelepiped's volume. The scalar rule for an $n \times n$ matrix is $\det(\lambda A) = \lambda^n \det A$.

A two-variable system is an intersection problem

The system $x + y = 3$ and $x - y = 1$ represents two lines. Their intersection is $(2, 1)$, so the unique solution is $x = 2, y = 1$. A parallel distinct second line would give no solution; the same line repeated would give infinitely many.

The solution is the common point of two lines



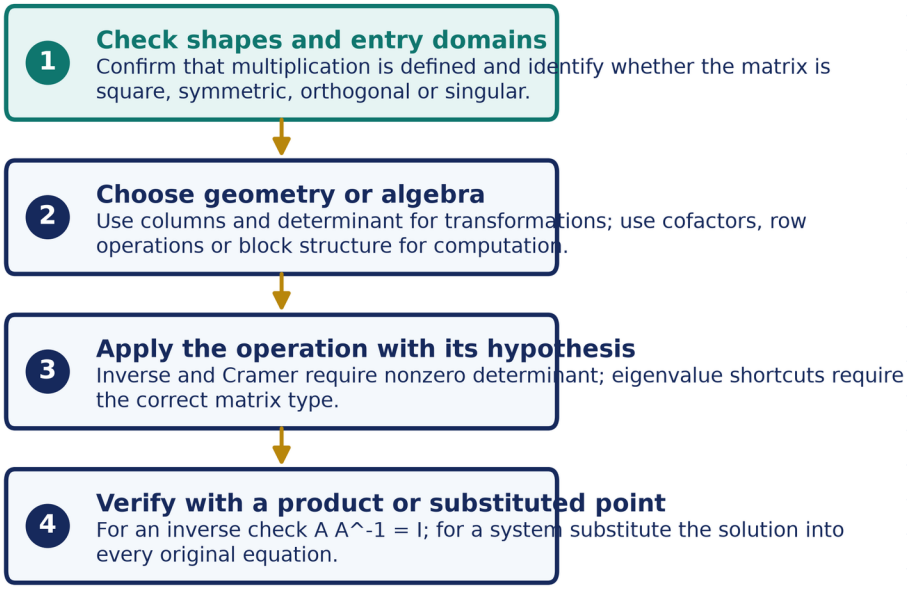
READ THE FIGURE

The determinant of the coefficient matrix is $1*(-1) - 1*1 = -2$, which is nonzero. This algebraic condition matches the geometric fact that the lines have one intersection.

Cramer's rule and the inverse method require the appropriate nonzero determinant. Row operations can still solve singular systems, but the answer may be inconsistency or a family rather than a unique point.

Matrix strategy: structure before arithmetic

A reusable matrix workflow



READ THE FIGURE

Matrix calculations become shorter when the structure is recognised first. Verification should use the original matrix or system, not an intermediate row-reduced form alone.

WORKED EXAMPLE

Verify an inverse instead of trusting one

For $A=\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\det A = 2$ and $A^{-1}=\begin{pmatrix} 1/2 & -1/2 \\ 0 & 1 \end{pmatrix}$. Multiplication in either order gives I.

This check catches both cofactor transposition errors and a missing determinant factor.

Cayley–Hamilton — Reduce Every Power

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 01 / 03

Characteristic polynomial

For square A, define $p(\lambda) = \det(\lambda I - A)$. Eigenvalues are roots because $(A - \lambda I)v = 0$ has a nonzero vector exactly when determinant vanishes. For 2×2:

$$p(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

KEY RESULT

Cayley–Hamilton theorem

Every square matrix satisfies its characteristic polynomial: $p(A) = 0$. For order 2, $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. Therefore every higher power reduces to a linear combination of I and A.

WORKED EXAMPLE

Power reduction

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, trace 5, determinant -2, so $A^2 - 5A - 2I = 0$. Then $A^2 = 5A + 2I$ and $A^3 = 27A + 10I$.

Eigenvalue invariants and similarity

$B = P^{-1}AP$ is a change of basis (similarity), when P invertible. Then $\det(\lambda I - B) = \det(\lambda I - A)$; hence trace and determinant, and the eigenvalue multiset, are unchanged. Distinct eigenvalues imply diagonalizability over a field containing all eigenvalues (for real matrices, at least over $\mathbb{C}$); repeated eigenvalues alone do not.

COMMON TRAP

Avoid an invalid substitution

Cayley–Hamilton lets us reduce powers of one fixed square matrix. It does not permit replacing a scalar variable by a matrix in an unrelated identity unless products commute and the polynomial identity applies.

LESSON 02 / 03

Hadamard's volume bound

For columns $c_1, \dots, c_n$ of A, $|\det A| \le \prod_j \|c_j\|$. Volume cannot exceed the rectangular box with those edge lengths; equality holds exactly when nonzero columns are pairwise orthogonal (zero-column case gives both sides zero).

SHORT DERIVATION

Two-dimensional identity

For columns $(a,c)^T, (b,d)^T$, $(ad-bc)^2 = (a^2+c^2)(b^2+d^2) - (ab+cd)^2$. Thus $|ad-bc|$ is at most the product of their lengths; equality iff dot product is zero.

Rank-one determinant lemma

For invertible A and compatible column vectors u, v:

$$\det(A + uv^T) = \det A(1 + v^T A^{-1} u).$$

A high-dimensional determinant update reduces to one scalar. Equivalent core identity: $\det(I + XY) = \det(I + YX)$ for compatible rectangular X, Y.

WORKED EXAMPLE

Check one update

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $u = (1,2)^T$, $v = (3,4)^T$. Directly $\det(A + uv^T) = -6$; the lemma gives $(-2)(1+2) = -6$.

Vandermonde determinant

$$\det \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

It vanishes iff two nodes coincide, exactly when quadratic interpolation loses uniqueness. General form: $\prod_{i<j}(x_j - x_i)$.

COMMON TRAP

State the needed hypotheses

The matrix determinant lemma uses A^{-1}, so A must be invertible. The Vandermonde product depends on the stated column order $1, x, x^2$ and row order x_1, x_2, x_3; reordering nodes changes sign accordingly.

Trace, Rank–Nullity, Block Structure

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 03 / 03

Trace: cyclic, not commutative

For square matrices A, B of the same order, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ since both equal $\sum_{i,j} a_{ij} b_{ji}$. This does **not** say $AB = BA$. More generally $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$ when dimensions permit cyclic products.

Rank–nullity

For linear map $A : \mathbb{R}^n \to \mathbb{R}^m$, $\operatorname{rank} A + \operatorname{nullity} A = n$. Row reduction exposes pivot variables and free variables; the null space is the solution space of $Ax = 0$.

WORKED EXAMPLE

Dimension count

If a 3×4 matrix has rank 2, then nullity is $4 - 2 = 2$: the homogeneous system has a two-dimensional family of solutions.

Block determinants

For square diagonal blocks A, D:

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D.$$

If A is invertible (and block sizes compatible), Schur complement gives

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B).$$

INTUITION

Eliminate the lower-left block

Multiply on the left by the block triangular matrix $\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$, whose determinant is 1. The product has lower-left block zero and lower-right block $D - CA^{-1}B$.

WORKED EXAMPLE

Schur complement in one calculation

Let $A = I_2$, $B = (1, 2)^T$, $C = (3, 4)$, $D = (10)$. Then the Schur complement is $10 - (3, 4)(1, 2)^T = 10 - 11 = -1$, so the block determinant is -1. Directly,

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 3 & 4 & 10 \end{pmatrix} = -1.$$

Boundary reminder

Eigenvalues, Cayley–Hamilton, rank–nullity, Hadamard, Vandermonde and Schur complements are useful Olympiad extensions; they are not being represented as explicit JEE syllabus requirements. Matrix algebra can still arise inside a JEE question when supplied or derivable.